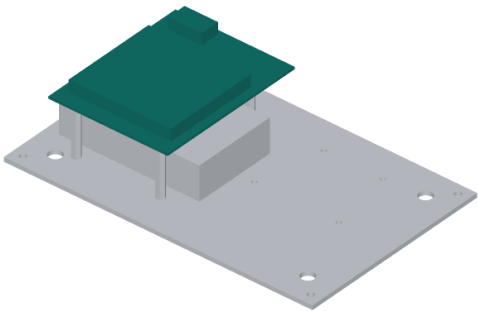


REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC
P2	Board lengthened to 108 mm for the coil economizer	2026-10-02	AC



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Cut 85 x 108 mm from 1.6 mm FR4 perforated board.

Four 3.2 mm holes for the standoffs: 15 and 75 mm from the edge nearest the far end, 6 and 82 mm from the socket-side edge.

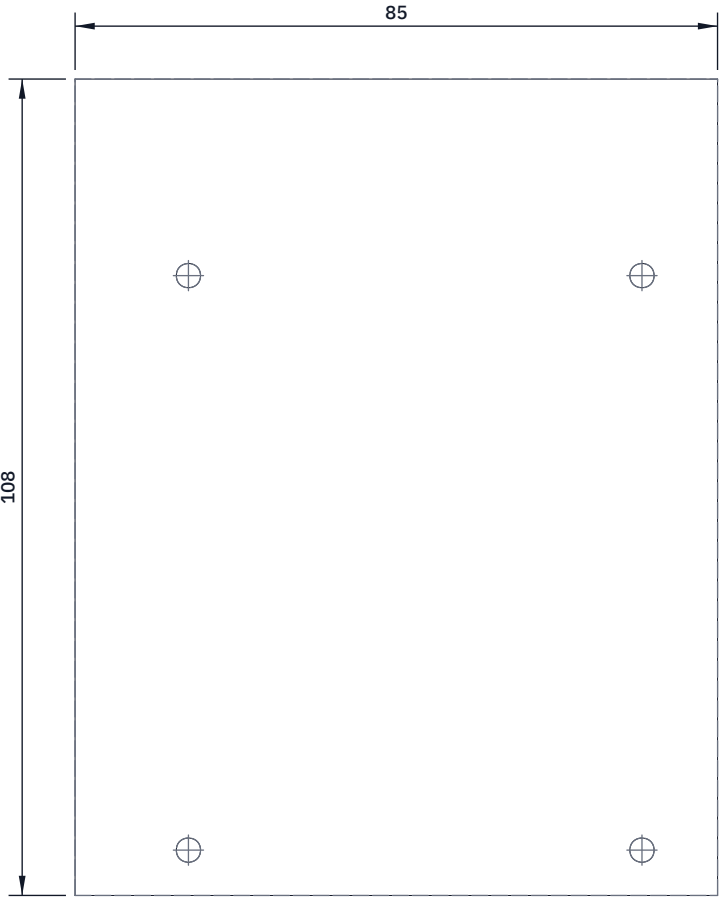
The prototype supervisor is built from modules on this board (see the wiring picture): microcontroller board with two CAN controllers, two CAN transceivers, 18 to 75 V in, 12 V out buck, coil switch, precharge switch, input conditioning for the speed, brake and e-stop B signals, and a USB socket for the fault log.

Coil economizer module (24 x 14 x 8): in the extra strip along the edge away from the sockets, 9 to 33 from the far end, 91 to 105 from the socket-side edge.

Keep all modules inside a 73 x 68 mm area in the middle so the modules clear the lid by 10 mm and the four standoff screws.

Fit: four M3 screws into 35 mm hex standoffs; the board sits over the controller with 10 mm of air between them.

Check: board flat; no solder tail longer than 2 mm underneath.



TOP VIEW

Scale 1:1; looking down (along -Z)



FRONT VIEW

Scale 1:1; looking at the front (along +Y)



RIGHT VIEW

Scale 1:1; looking at the right side (along -X)

MotionCore supervisor carrier board: making sketch

DRAWING NO. MTC-DWG-103		REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:1	UNITS mm	PROJECTION 	DATE 2026-10-01	DRAWN Amish Chadha
MATERIAL / NOTES FR4 perforated board 1.6 mm, 2.54 mm pitch				
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